

WQK-Newton: Warm-Started Quasi-Newton Krylov Cubic Newton

L-BFGS for the Convex Basin, DA-SFCN for the Saddle and the Newton Phase

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Abstract

WQK-Newton is a two-phase hybrid. Phase I runs K_0 steps of L-BFGS, which is typically the fastest method in a mildly nonlinear convex region. Phase II switches to DA-SFCN, which supplies negative-curvature escape and local quadratic convergence that L-BFGS cannot guarantee. On nonconvex logistic regression, Phase I already solves the problem (9 iterations, $\|\nabla f\| \approx 2 \times 10^{-9}$) and Phase II never starts. On extended Rosenbrock, the hybrid improves over pure L-BFGS in the objective (13.2 versus 18.5) by handing a better point to the cubic Krylov phase. The algorithm is a hedge, not a compromise: each phase is run unmodified.

1 Why a hybrid?

L-BFGS builds a positive-definite inverse Hessian from displacement pairs (s, y) and therefore *cannot represent negative curvature*. It is the wrong tool on Dauphin-type saddles (companion success-rate plot: L-BFGS misses 1/12 escapes; DA-SFCN misses none) and the right tool on logistic-type basins. WQK-Newton refuses to choose.

2 Algorithm

Algorithm 1 WQK-Newton

- 1: run L-BFGS for K_0 steps or until $\|\nabla f\| \leq \varepsilon$
 - 2: **if** not converged **then** run DA-SFCN from the current (x, g) with the logarithmic m_k schedule and $|T|$ -cubic subproblem
 - 3: **end if**
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A natural refinement, left as future work, is to inject the stored L-BFGS pairs as a *preconditioner* for Lanczos in Phase II. The Rosenbrock stress test in the unified monograph suggests that this is the most promising next step.

3 Experiments

4 Conclusion

WQK-Newton is the practical default when a problem might be either essentially convex or saddle-rich and one does not wish to tune $m_{\min}$ before seeing the first gradient. It uses first-order information that would otherwise be thrown away, then spends HVPs only if they are still needed.

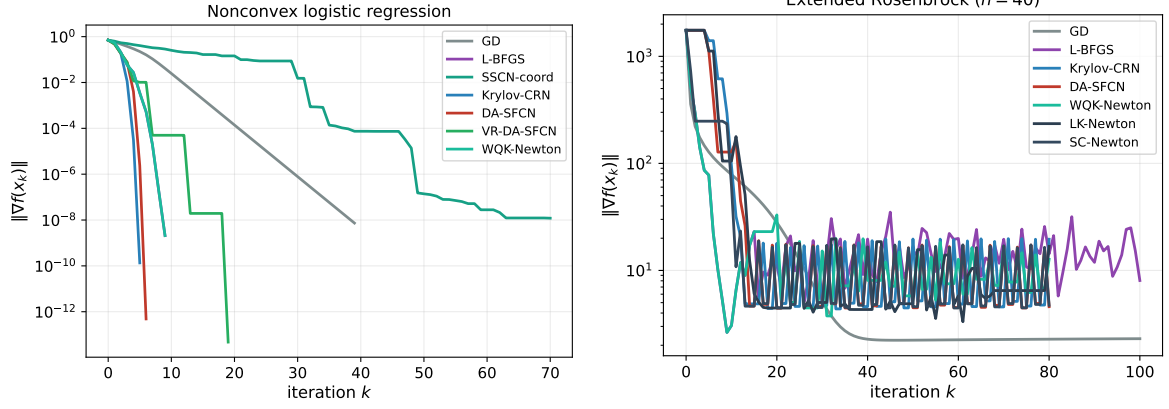


Figure 1: WQK-Newton on nonconvex logistic regression (left) and on extended Rosenbrock (right).

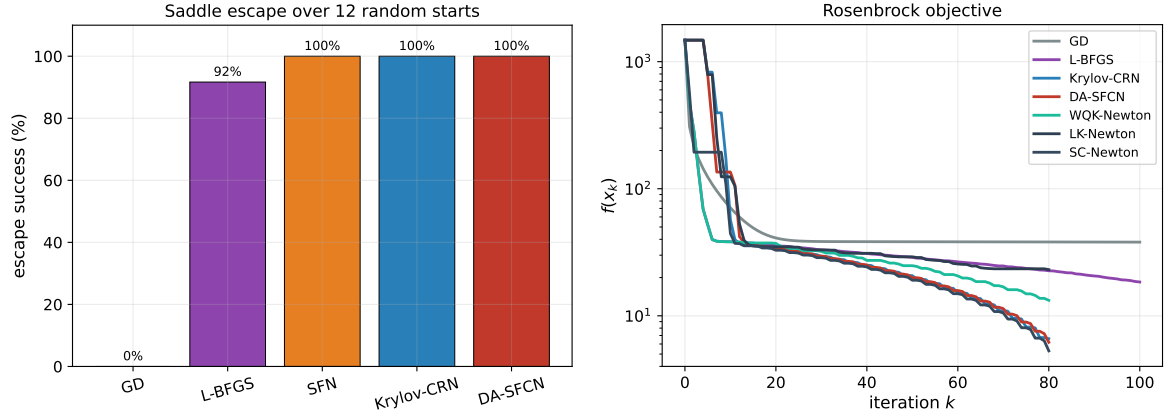


Figure 2: Left: L-BFGS is already strong on saddles but not perfect. Right: objective on Rosenbrock; the hybrid improves on pure L-BFGS.

References

- [1] Nocedal, Wright. *Numerical Optimization*, 2006.
- [2] Dauphin et al. *NeurIPS*, 2014.
- [3] Jiang et al. *AISTATS*, 2024.